20 Solve algebraically the simultaneous equations

$$x^2 - 4y^2 = 9$$
$$3x + 4y = 7$$

$3x + 4y = 7 \rightarrow x = (7 - 4y)/3.$

1 mark
Rearrange one equation.

Substitute into $x^2 - 4y^2 = 9$:

$(7 - 4y)^2/9 - 4y^2 = 9$

1 mark
Correct substitution.

$49 - 56y - 20y^2 = 81$

$5y^2 + 14y + 8 = 0$

1 mark
Quadratic $5y^2+14y+8=0$.

$(5y + 4)(y + 2) = 0$

$y = -4/5$ or $y = -2$.

1 mark
Factor and solve y.

$x = 17/5$ when $y = -4/5$; $x = 5$ when $y = -2$.

1 mark
Paired (17/5,−4/5),(5,−2).

(17/5, -4/5), (5, -2)

(Total for Question 20 is 5 marks)

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